CSC 6740 **Data mining** **D**ue by: Oct. 11th

Name:

Assignment 3

Q1: The Apriori algorithm makes use of *prior knowledge* of subset support properties. (5 points)

a) Explain that the (relative) support of any nonempty subset s' of frequent itemset s must be at least as great as the (relative) support of s.

b) Given frequent itemset L and subset s of L, prove that the confidence of the rule "$s' \Rightarrow (L - s')$" cannot be more than the confidence of "$s \Rightarrow (L - s)$", where s' is a subset of s.

Q2: A database has 5 transactions. Let *min sup* = 60% and *min conf* = 80%. To find the frequent patterns (including 1-itemsets)

TID	*items bought*
T100	{M, O, N, K, E, Y}
T200	{D, O, N, K, E, Y }
T300	{M, A, K, E}
T400	{M, U, C, K, Y}
T500	{C, O, K, I, E}

(a) Find all frequent itemsets using Apriori. Please do **NOT** call the functions from existing package or library. Do it step by step by hands or by your own code. Grades are given based on intermediate results from each step. (20 points)

(b) Find all frequent itemsets using FP-growth. Please do **NOT** call the functions from existing package or library. Do it step by step by hands or by your own code. Grades are given based on intermediate results from each step. (20 points)

(c) Explain why FP-Growth is bounded by exactly two database scans regardless of the length of the longest frequent pattern, while Apriori requires k database scans to discover a frequent k-itemset, excluding checking whether K+1 itemset frequent (5 points).

(d) List all of the ***strong*** association rules (with support *s* and confidence *c satisfying the min requirements*) matching **the following rule**, where *X* is a variable representing a customer, and *item_i* denotes variables representing items (e.g., *"A"*, *"B"*,"M" i.e., if any customer buys A and B, then the customer will buy M, Three Items are involved) (5 points)

$\forall x \in transaction,\ buys(X, \mathbf{item_1}) \wedge_{and} buys(X, \mathbf{item_2}) \Rightarrow buys(X, \mathbf{item_3})$

Q3: Give a short example to show that items in a strong association rule (Ex: min_support=10%, min_confidence=50%) may actually be *negatively correlated*. To detect and remove negatively associated rules, what measure(s) you can use (5 points)

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Q4: Coding exercise(40 points): Using the Data File: retail_200_expanded.csv in iCollege, you will compare a Apriori with FP-Growth for frequent pattern mining;

- Minimum Support Threshold (minsup): 25%
- Minimum Confidence Threshold (minconf): 80%

Frequent Itemsets extraction (20 pts):

a) Execute frequent itemset extraction via the Apriori module at a strict 25% support threshold.
b) Execute frequent itemset extraction via the FP-Growth module at the same 25% support threshold.
c) Programmatically assert that both approaches return identical frequent itemset groupings. Print a table listing all discovered frequent itemsets of length >= 2 alongside their absolute transaction frequencies.

Rule Resolution Engine (20 pts):

a) Generate all non-empty directional association rules matching a confidence metric of >= 80%. For each qualifying rule, compute and display: Support, Confidence, and Lift. Sort your output table in descending order of Lift. Are they all meaningful rules based on Life?